Supplementary Information for
An attractor account of numerosity perception

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This document accompanies the article *An attractor account of numerosity perception*. Supplementary Material 1 details the trial-level fitting; Supplementary Material 2 reports the full model comparison and the constrained cross-duration fits; Supplementary Material 3 the parameter recovery; Supplementary Material 4 the resolution-limit prediction and its robustness; and Supplementary Material 5 the exploratory developmental-dyscalculia reanalysis.

Supplementary Material 1: Trial-level fitting

Data. We reconstruct one row per trial from the openly available data of Cheyette and Piantadosi (2020), grouping by participant and trial and keeping the presented numerosity, the typed response, and the response time (converted to seconds). We use the five exposure durations (40, 80, 160, 320, 640 ms), true numerosities $n = 1$–15, giving ≈10,600 trials per duration and ≈53,000 in total. No trials are aggregated: the likelihood is evaluated over the individual integer reports.

Response likelihood. For each model and duration the response probability of Eq. (4) marginalises the readout over the sensory channel $y_0 \sim \mathcal{N}(n, \sigma(n))$. The integral is computed *deterministically*, so the objective is free of Monte-Carlo noise. Gauss–Hermite quadrature is inappropriate here: for the attractor models the clean-up map $y_0 \mapsto x^*$ is near-discontinuous (evidence snaps to basins), and Gauss–Hermite carries a 1.7–4% error even at 241 nodes. We therefore use a uniform standard-normal grid with Gaussian weights for *every* model, so the comparison is on identical footing. Against high-sample Monte-Carlo estimates, a grid of 801 nodes matches the integral to $\leq 2 \times 10^{-3}$ (mean 1.5×10^{-4}). The optimiser uses a cheaper grid (201 nodes); the reported log-likelihoods are recomputed at the fitted parameters on a fine grid (1601 nodes).

Lapse channel. Typed responses run up to 99, and a few per cent of guesses fall outside the integer-well support (3.9–6.1% across durations). We never discard them: the final response probability is $p_{\text{final}}(r \mid n) = (1 - \ell)\, p(r \mid n) + \ell \,/\, R_{\max}$ with $R_{\max} = 99$, a uniform contamination that places positive mass everywhere. The lapse ℓ is fitted per duration under an identical convention across all models and is counted in AIC/BIC; it is stable across models (≈0.06–0.08).

Response-time and coupled channels. Response time is modelled as log-normal, $\log \mathrm{rt} \sim \mathcal{N}(\log(t_0 + m(n)), \sigma_{\mathrm{rt}})$, with the task-fixed base $t_0 = 1.30$ s. The metric $m(n)$ is the expected posterior entropy ($\kappa\, \mathbb{E}[H]$, available to every model), the expected convergence time ($\tau\, \mathbb{E}[T]$, attractor only), or their sum (the hybrid; Eq. 3). Because the log-normal maximum-likelihood fit targets the mean of $\log \mathrm{rt}$, i.e. the geometric-mean RT, all reported R^2 values compare the prediction against the geometric mean rather than the median. The coupled $p(r, \mathrm{rt} \mid n)$ integrates response and RT over the same latent nodes. Co-optimising all channels together (warm-started from the two-stage fit) leaves the model ranking unchanged and barely moves the landscape, so we treat RT as a downstream cross-prediction.

Optimisation. The deterministic objective is minimised by `scipy` differential evolution (Storn and Price 1997) followed by a Nelder–Mead polish, with fixed seeds. The attractor models require a larger optimiser budget than the smooth static readouts (a low-budget fit understates them); re-fitting at a higher budget with a different seed changes the reported log-likelihoods by less than one part in 10^3, so the fits are converged and the model ordering is not an optimisation artefact.

Supplementary Material 2: Model comparison and constrained fits

Table S1 gives the full trial-level response comparison with each duration fitted separately; Table S3 gives the complementary joint fit, one landscape shared across all durations. Every model is fitted under the identical convention (per-duration parameters plus a per-duration lapse, summed over the five durations). The super-linear-width Gaussian attractor is the best account under both fits, and the ranking separates the contributions of its two ingredients: the clean-up dynamics and the width scaling.

Turning the clean-up dynamics on is a zero-parameter change (the dynamics are deterministic given the landscape),

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Table S1. Trial-level response model comparison. Categorical information criteria over the $\sim$53,000 individual responses, summed over the five exposure durations, relative to the best model. k is the total free-parameter count (per-duration mechanism parameters plus lapse). "Attractor" includes the clean-up dynamics; "readout" is the matched static control at the same k. The width-model rows carry one parameter more than the Weber/ANS and normative benchmarks (the free width exponent); at this sample size BIC charges $\approx$9 per parameter against AIC's 2, yet the ranking and margins barely change, so the advantage is not a parameter-count artefact.

Model	k	ΔAIC	ΔBIC
Super-linear attractor (free β)	**25**	**0**	**0**
Super-linear attractor (power-law n^p)	25	165	165
Super-linear readout (free β, static)	25	1256	1256
Power-law readout (static)	25	1491	1491
Weber/ANS ($\sigma_n \propto n$)	20	4435	4398
Normative log-prior proxy	20	7666	7629

Table S2. Fitted parameters of the winning model across exposure durations. σ_c and β set the well-width law $\sigma_n = (\sigma_c/f_n)^\beta$; s_0 and λ_0 the sensory-noise profile $\sigma(n)$; ℓ the lapse. The 40 ms flash is noise-dominated (large s_0, extreme β); from 80 ms the parameters vary smoothly.

Duration	σ_c	s_0	λ_0	β	ℓ
40 ms	0.085	0.023	2.91	3.42	0.080
80 ms	0.054	0.0056	3.03	1.81	0.064
160 ms	0.047	0.0023	3.25	2.02	0.059
320 ms	0.045	0.00047	3.76	2.19	0.079
640 ms	0.041	0.00036	3.67	2.15	0.072

so the attractor and its static twin occupy the identical parameter space. Within the super-linear family this flips the fit by $\Delta\text{AIC} = 1256$ in favour of clean-up, and within the power-law family by 1326, so the dynamics do explanatory work rather than add flexibility. (Per duration, clean-up helps strongly at 40–160 ms and is roughly neutral at 320–640 ms, where the attractor has largely settled during the exposure itself; main-text Fig. 2b.)

The width exponent is a free parameter of the model rather than a fixed value. It is estimated firmly super-linear at every exposure except the noise-dominated 40 ms flash (Table S2). This super-linearity is what gives the wells their sharp-to-broad profile: narrow and well-separated for small numbers, wide and overlapping for large ones. Granting the Weber/ANS code the same flexibility does not close the gap: freeing its width exponent (the power-law readout, also $k = 25$) leaves it 1491 behind the winning attractor, and the freed exponent itself settles super-linear rather than at the Weber value $\sigma_n \propto n$. At equal parameter count, then, a magnitude code still loses, and it does not even prefer the Weber scaling.

The comparison in Table S1 fits each duration separately. Because the landscape is a property of the number system rather than of the display, we also fit a single landscape (σ_c and the width exponent) shared across all five durations, leaving only the sensory channel and lapse free per duration (Table S3; main-text Fig. 2a). The attractor remains the best account. With one shared landscape (17 parameters) it still beats the fully free per-duration Weber/ANS code (20 parameters) by $\Delta\text{AIC} = 905$ ($\Delta\text{BIC} = 932$) and the normative proxy by 4136, and the clean-up dynamics still improve the fit at equal parameters by $\Delta\text{AIC} = 781$. Forcing

Table S3. Joint fit: one landscape shared across the five durations. Categorical information criteria over the $\sim$53,000 responses relative to the best model. The four well models share their landscape (σ_c and the width exponent) across durations, with the sensory channel and lapse free per duration ($k = 17$). The Weber/ANS and normative benchmarks have no landscape separable from their sensory noise, so they keep their fully free per-duration fit ($k = 20$); the attractor's single landscape wins even against that extra freedom.

Model	k	ΔAIC	ΔBIC
Super-linear attractor (free β)	**17**	**0**	**0**
Super-linear attractor (power-law n^p)	17	149	149
Super-linear readout (free β, static)	17	781	781
Power-law readout (static)	17	789	789
Weber/ANS ($\sigma_n \propto n$)	20	905	932
Normative log-prior proxy	20	4136	4163

the landscape to be duration-independent costs $\Delta\text{AIC} = 3529$ relative to the separate fits ($\Delta\text{BIC} = 3458$), and the cost is concentrated in σ_c (sharing only σ_c costs 3398, only β costs 990). The landscape therefore does change with exposure, mostly by sharpening: this is the resolution growth that drives n_{crit} (Supplementary Material 4). We read this as one mechanism whose resolution improves with viewing time, not a different mechanism per duration.

In the winning model the number frequency f_n enters only through the well width. We also tested adding a frequency prior at the readout stage ($p(n \mid x^*) \propto f_n \exp g_n(x^*)$), a zero-parameter change. It does not robustly improve the fit (summed $\Delta\text{AIC} = -125$ but inconsistent in sign across durations, helping at 80–160 ms and hurting at 40, 320, and 640 ms), so we retain the flat readout and let the frequency dependence live entirely in the well width.

When the response and response-time channels are fitted together as a single coupled likelihood, rather than treating the response time as a downstream cross-prediction, the conclusions are unchanged: the winning attractor with the dynamics-based hybrid RT link remains best, ahead of the Weber/ANS code by $\Delta\text{AIC} = 3913$. Within the winner, the convergence-time (hybrid) link improves this coupled fit over the static entropy proxy by $\Delta\text{AIC} = 784$, so the attractor dynamics carry the response-time channel even when it is fitted together with the response distribution.

Supplementary Material 3: Parameter recovery

To confirm that the fitted quantities are identifiable at the trial level, we ran a parametric recovery: for each duration we generated 50 synthetic datasets from parameters jittered around the fitted solution, sampled with the empirical per-numerosity trial counts and the lapse channel, and refitted them with the exact headline pipeline (250 refits in total). Table S4 reports the recovery of each parameter and of the derived quantities the paper interprets; Figure S1 shows the scatter and the error-correlation structure.

The width exponent β and the derived resolution limit n_{crit}, the two quantities that carry the paper's argument, are recovered accurately ($R^2 = 0.89$ and 0.996). The only strong trade-off in the error-correlation matrix is between the two sensory parameters s_0 and λ_0 ($\rho = -0.97$); so we

Table S4. Trial-level parameter recovery of the winning model. Identity R^2 and Spearman ρ between generating and recovered values across the 250 synthetic datasets. Scale parameters are compared on the log scale.

Quantity	R^2	Spearman ρ
σ_c (well scale)	0.99	1.00
β (width exponent)	0.89	0.95
s_0 (sensory scale)	0.95	0.96
λ_0 (sensory exponent)	0.83	0.94
ℓ (lapse)	0.97	0.98
n_{crit} (derived)	1.00	1.00
σ_3 (well width at $n = 3$)	0.98	1.00
$\sigma(n = 5)$ (sensory sd)	0.99	1.00

never interpret them separately, only through the combined sensory-noise profile $\sigma(n) = \sqrt{s_{00}^2 + s_0 n^{\lambda_0}}$, which is itself recovered accurately. All landscape quantities (σ_c, β, n_{crit}) are cleanly identified.

Supplementary Material 4: The resolution limit n_{crit}

The critical numerosity is defined by adjacent-well discriminability $d' = 1$, i.e. $\sigma_n = 1$ (well standard deviation equal to unit spacing) (Macmillan and Creelman 2004). Solving $\sigma_n = (\sigma_c / f_n)^\beta = 1$ for the fitted parameters gives $n_{\text{crit}} \approx 4.6, 7.5, 8.7, 9.2, 10.0$ for 40–640 ms. The crossing lies inside the fitted range at every duration and grows monotonically with exposure (main-text Fig. 2e).

Because the crossing falls inside the fitted window, the estimate could in principle be shaped by the large-n tail rather than by the small-number range itself. It is not: refitting on $n \leq 10$ alone returns nearly the same width exponent and n_{crit} as the full-range fit for 80–640 ms (e.g. $\beta = 1.88$ vs. 1.81 at 80 ms, 2.09 vs. 2.19 at 320 ms; n_{crit} within 0.3 at all four), and only the noise-dominated 40 ms flash shifts (Fig. S2). Within $n \leq 10$ the attractor still comes out ahead of its static twin ($\Delta\text{AIC} = 172$) and of the Weber/ANS code (2043). The exact code is thus a property of the small-number range, not an artefact of fitting the tail.

The response-time channel points to the same boundary. Within the exact range ($n < n_{\text{crit}}$) the model tracks the geometric-mean response time closely; above it the monotone deliberation-time prediction keeps rising while the empirical RT plateaus, because participants have left the exact regime and are guessing approximately (Fig. S3). The onset of the empirical RT plateau thus coincides with n_{crit}, even though the latter is derived from the accuracy channel alone. We therefore fit the latent-to-seconds RT link on all n and report the within-domain agreement; truncating the fit to $n < n_{\text{crit}}$ is counter-productive, since at short durations the cut falls just before saturation and the truncated fit overshoots.

The resolution limit also leaves a visible mark in exact accuracy. The landscape is smooth by construction (the width law and the sensory-noise profile are smooth, monotone functions of n), yet the predicted proportion of exact responses shows a small shoulder at the resolution limit, visible in the main-text co-fit (main-text Fig. 3b). This is not extra fitted flexibility but the discrete clean-up basins on the fixed integer lattice: around n_{crit} the last wells that still resolve snap noisy evidence back onto the exact answer and prop exact accuracy up at the edge of the exact range. The shoulder vanishes when the clean-up is switched off at identical parameters (Fig. S4). The shoulder thus tracks n_{crit} across durations and is a signature of the mechanism rather than of additional degrees of freedom.

Last, we asked whether the fitted model also reproduces the effective Weber fraction. We computed the coefficient of variation (CoV) of the fitted response distribution, $(1 - \ell)\, p(r \mid n) + \ell / R_{\max}$ with the lapse included, per numerosity and exposure. We restricted it to the in-support range, so that it is the like-for-like counterpart of the empirical estimates, and overlaid the empirical CoV (Fig. S5). The model tracks the data across the whole range at every exposure. The CoV is elevated at the smallest numerosities (where the ratio sd/mean is inflated by the small denominator, in model and data alike) and settles to a roughly constant plateau across the estimation range ($n \geq n_{\text{crit}}$). A power-law fit $w_{\text{eff}} \propto n^\gamma$ over the estimation range gives a pooled exponent $\gamma = -0.21$ for the model and -0.17 for the data: approximately scalar variability, close to the Weber value $\gamma = 0$. The same landscape thus reproduces the numerosity variability, and its slope, across the whole range.

Supplementary Material 5: Exploratory reanalysis of developmental dyscalculia

As an exploratory extension we applied the winning model to the exact-enumeration data of Decarli et al. (2020): accuracy and response time per subject and numerosity for 32 children with developmental dyscalculia (DD) and 32 age-matched controls, using numerosities $n = 1$–7 (dropping $n = 8$ exactly as the original authors do, for the guessing end-effect they report). The dot arrays were shown for a fixed exposure of 500 ms, so all quantities in this section, including n_{crit}, refer to that exposure. The model fits both groups well (Fig. S6a,b) and again prefers a super-linear width ($\beta \approx 1.5$–1.6), replicating the main dataset. The fitted $n_{\text{crit}}(\sigma_n = 1) \approx 8$ falls slightly below the adult values of 9.2–10.0 at 320–640 ms (Supplementary Material 4), as one would expect for children. Although these fits on 64 subjects can only give us very preliminary intuitions, we can make two tentative observations.

First, the decline of accuracy with numerosity is better explained by a loss of resolution than by sensory noise. With an exhaustive per-subject grid search and a fixed common lapse, we fitted two single-mechanism variants of the model to each child's accuracy curve: a resolution-only variant, in which sensory noise is switched off and the decline is carried entirely by the merging of wells, and a noise-only variant, in which the wells never merge and the decline is carried entirely by sensory noise. The resolution-only variant fits better in $64/64$ subjects (mean SSE 0.010 vs. 0.079, Wilcoxon $p = 3.5 \times 10^{-12}$; Fig. S6c). The decline of exact enumeration in this dataset therefore behaves as a loss of discrete resolution, which supports the attractor framework on independent data and constrains where a group difference can arise.

Second, the group difference falls on that same resolution parameter. Per subject, the resolution limit is shorter in DD

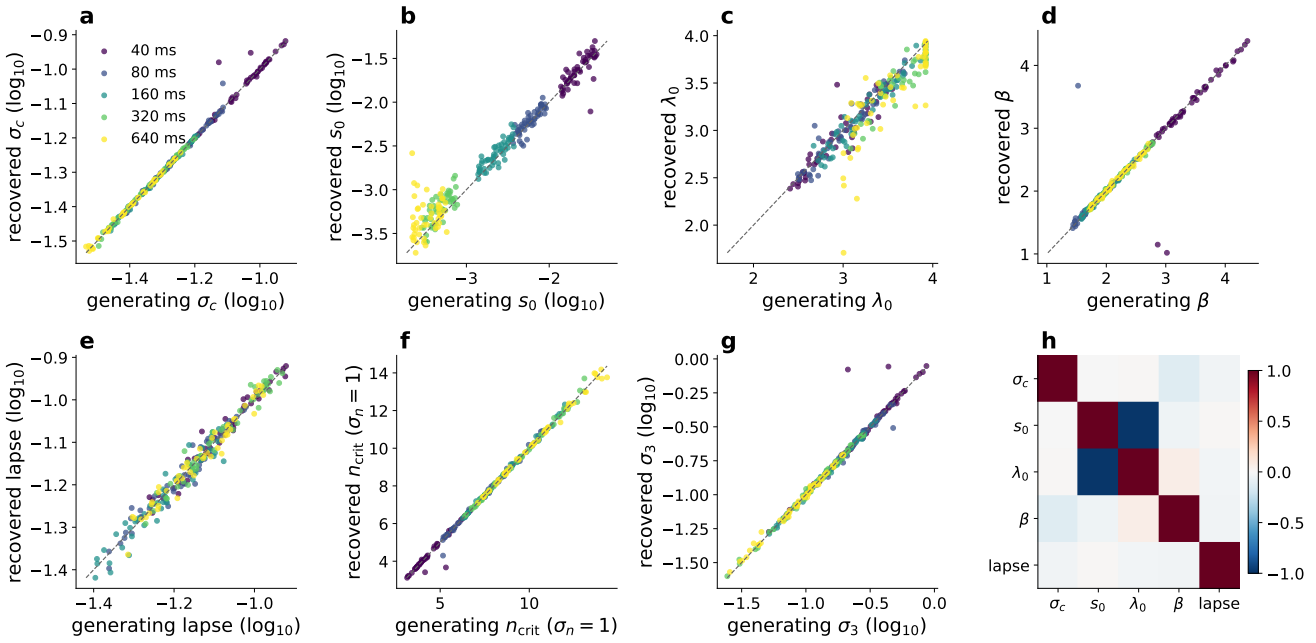


Figure S1. Parameter recovery of the winning model. Recovered versus generating values for the five fitted parameters and the derived n_{crit} and σ_3, coloured by exposure duration; dashed lines are identity. The bottom-right panel is the Spearman correlation matrix of recovery errors, showing the single strong s_0–λ_0 trade-off.

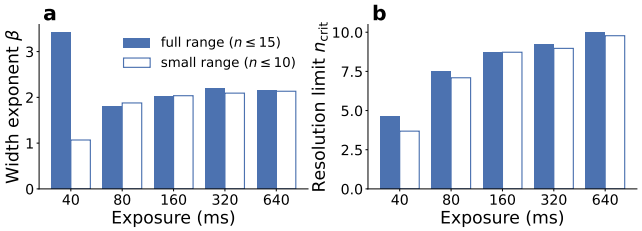


Figure S2. Small-range refit. The fitted width exponent β (**a**) and the resolution limit n_{crit} (**b**), from the full-range fit ($n \leq 15$, filled) and from a refit on the small range ($n \leq 10$, open). Both are essentially unchanged for 80–640 ms; only the noise-dominated 40 ms flash shifts. The discrete, exact behaviour is intrinsic to the small-number range.

($n_{\mathrm{crit}} = 8.2$ vs. 9.1 for controls; Mann–Whitney $p = 0.007$, $r_{\mathrm{rb}} = -0.39$ with the RT channel included; $p = 0.044$, $r_{\mathrm{rb}} = -0.29$ in the accuracy-only fits of Fig. S6d), whereas the mirror analysis that frees sensory noise instead shows only a weak trend ($p = 0.07$) and fits worse. In rank terms the n_{crit} difference is comparable to the raw accuracy deficit at $n = 4$–6 ($r_{\mathrm{rb}} = -0.33$): the shift of ≈ 0.9 numerosities looks small only because of the scale it is expressed on.

On this model, then, developmental dyscalculia is not a noisier perception of number but a coarser exact code: the discrete resolution runs out earlier, at the subitizing edge. The claim is mechanistic and falsifiable, although much of the statistical power comes from analysing subjects individually rather than from the model itself. It is also complementary to the account of Dolfi et al. (2024), who locate the dyscalculic deficit in a weaker number sense within an image-computable network. Both accounts place the deficit in the number code, and coupling the attractor to such a front end (which the present model sets aside) would allow a resolution deficit and a sensory-noise deficit to be separated within a single model. The analysis remains exploratory: the data are coarse (8 trials per cell), the localisation to resolution rests on the per-subject model comparison rather than on a full identification of the two mechanisms, and a small sensory-noise component may coexist.

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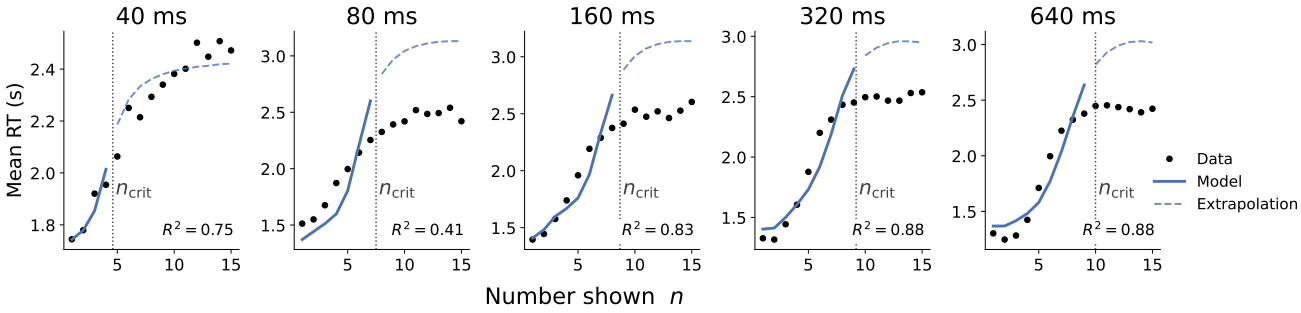


Figure S3. Response time within and beyond the exact range. Predicted geometric-mean RT (solid, $n < n_{\mathrm{crit}}$; dashed, extrapolated above) against the data. Within the exact range the model tracks RT; above n_{crit} it over-predicts, the signature of the switch to approximate responding. The RT plateau onset coincides with the independently derived n_{crit}.

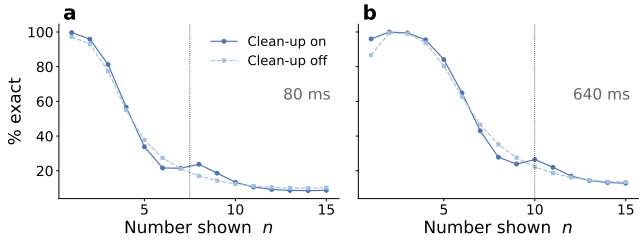


Figure S4. The exact-accuracy shoulder is a clean-up signature. Predicted proportion of exact responses at 80 ms (**a**) and 640 ms (**b**) with the clean-up dynamics on (solid) and off (dashed), at identical parameters; dotted verticals mark n_{crit}. The shoulder sits at n_{crit} and disappears without clean-up: it reflects the discrete basins on the integer lattice, not extra parameters.

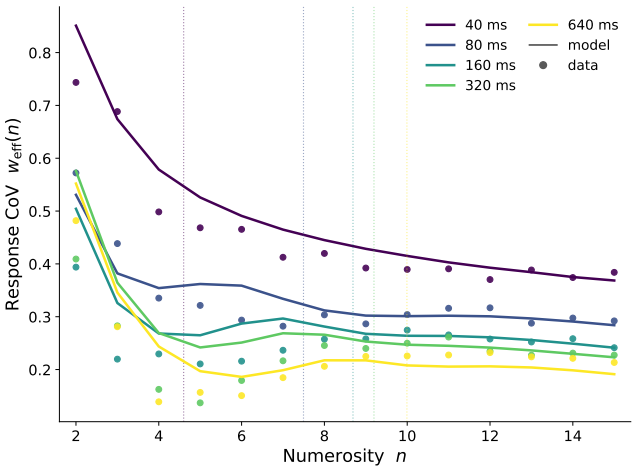


Figure S5. The effective Weber fraction. Coefficient of variation of the response distribution $w_{\mathrm{eff}}(n)$ per numerosity, for each exposure duration; solid lines are the model (the fitted response distribution, lapse included, restricted to in-support and renormalised), dots the data (in-support responses), and dotted verticals mark n_{crit}. The model tracks the data across the whole range; across the estimation range ($n \geq n_{\mathrm{crit}}$) the CoV settles to a roughly constant plateau, approximately scalar variability (Weber's law). The 40 ms flash is noise-dominated and has no exact range.

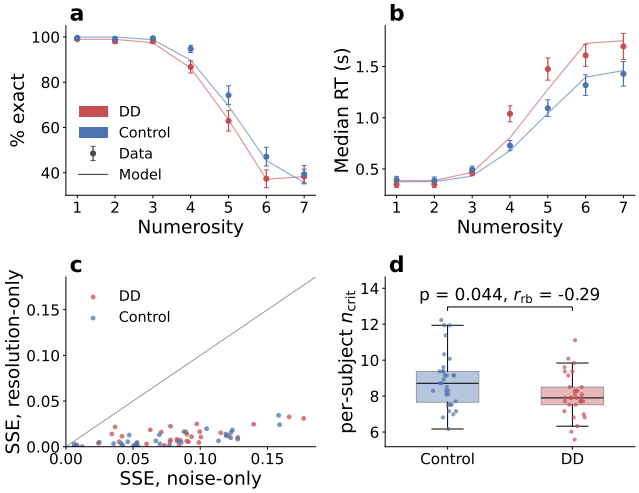


Figure S6. Developmental dyscalculia as a shorter exact number code. (a) Exact accuracy per numerosity (points, group mean ± SEM; lines, the group-level model fit). (b) Median response time, same conventions. (c) Per-subject accuracy SSE of the two single-mechanism fits: the resolution-only model fits better than the noise-only model in $64/64$ subjects (all points below the diagonal). (d) Per-subject resolution limit n_{crit} ($\sigma_n = 1$) from the resolution-only accuracy fits, at the task's 500 ms exposure. Boxes, median and interquartile range; whiskers, $1.5\times$ IQR; points, individual subjects; the bracket gives the Mann–Whitney p and the rank-biserial correlation r_{rb} (negative: DD resolve to fewer numerosities) for this accuracy-only analysis.